SONIC

An On-Device Sound Awareness System for Deaf and Hard-of-Hearing Users

Project Review

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Finalized Problem Statement

Critical Missed Cues



Smoke Alarms & Sirens



Doorbells & Phones



Appliance Beeps & Household Alerts

Limitations of Existing Systems

- Requires Cloud Connectivity
- Limited to Single Personalized Cues
- Inflexible in Diverse Acoustic Environments

Sonic proposes a privacy-oriented, on-device sound-awareness system that identifies general environmental sounds and user-taught household sounds, then provides timely tactile, visual, local notification, and BLE wearable alerts.

Objectives

Ambient Pipeline & Detection

- [x] - Develop an on-device ambient audio pipeline using 16 kHz mono PCM input.
- [x] - Detect 20 environmental sound categories arranged by urgency.

Personalization & Edge Storage

- [x] - Support user-taught sounds using few-shot learning from two to five samples.
- [x] - Reduce repeat alerts using a 10-second per-sound debounce mechanism.
- [x] - Store sound prototypes, history, feedback, thresholds, and settings locally in SQLite.

Accessible Alerting & Evaluation

- [x] - Provide tactile vibration, local notifications, visual flash alerts, and BLE wearable alerts.

- [ ] - Adapt personal-sound thresholds after three consecutive negative feedback ratings.
- [ ] - Evaluate model behavior, usability, and robustness in later project phases.

Literature Review

Reference Work	Core Focus	Architectural Insight
ProtoSound (Jain et al., 2022) & ACDNet (Gourisaria, 2021)	Scalable, real-time edge processing	Validates personalized, efficient edge-device DL pipelines.
AHEAD-DS / YAMNet+ & Nguyen et al., 2018	Assistive hearing contexts & wearable devices	Confirms viability of continuous wearable sound detection.
ESC-50, UrbanSound8K, AudioSet	Foundational environmental audio	Provides required baseline for general sound taxonomy.

Lumename Base-Paper Comparison

Lumename (Dao & Dao, 2025)

- Personalized single keyword/name detection
- MFCC + custom CNN from scratch
- Name-calling awareness only
- Haptic-visual alerts

SONIC (Proposed Architecture)

- Multi-class environmental + personal household matching
- YAMNet-based transfer learning + 128-D cosine similarity
- Broad two-tier design: universal categories & few-shot prototypes
- Tactile, local notification, flash, and BLE wearable alerts

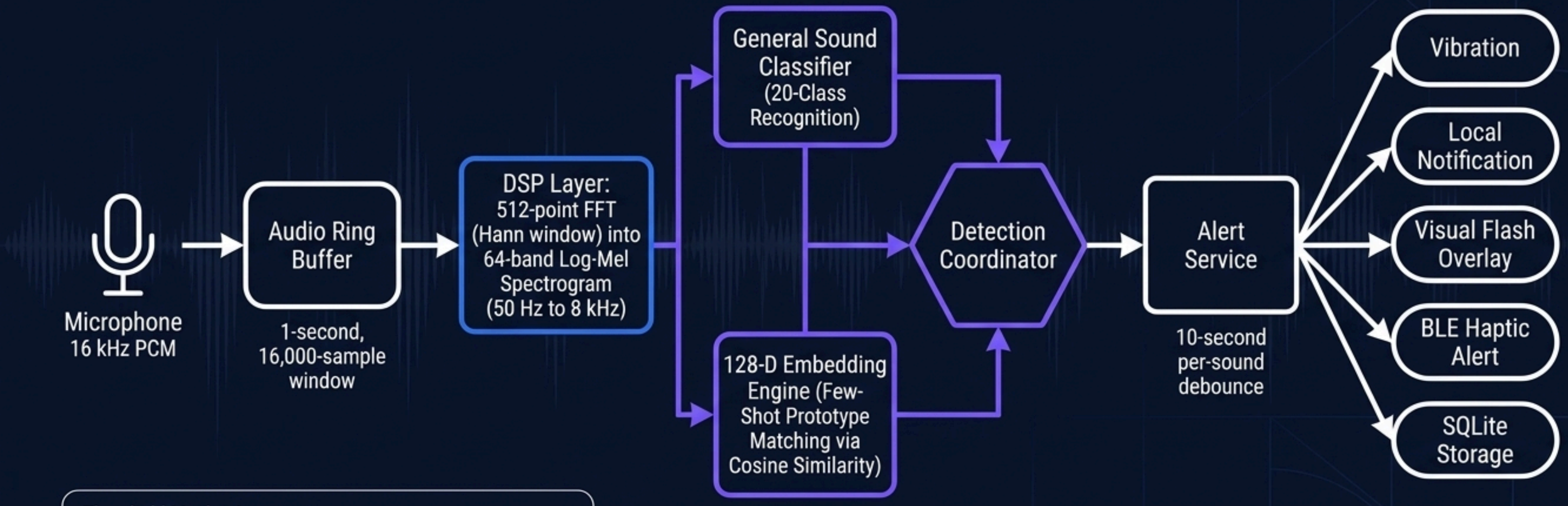
Research Gap



Combines universal environmental sound categories with local few-shot prototypes for personally meaningful sounds

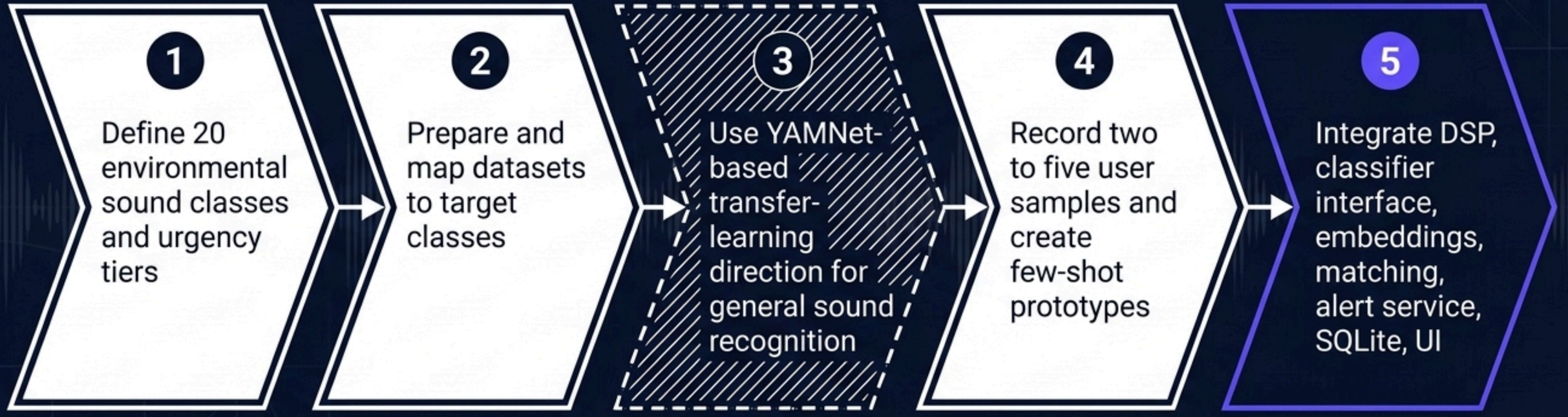
The proposed contribution is an on-device architecture integrating DSP, classification, prototype matching, adaptive feedback, multimodal alerts, and local persistence. Performance and usability benefits require future evaluation phases.

Proposed Architecture and System Design

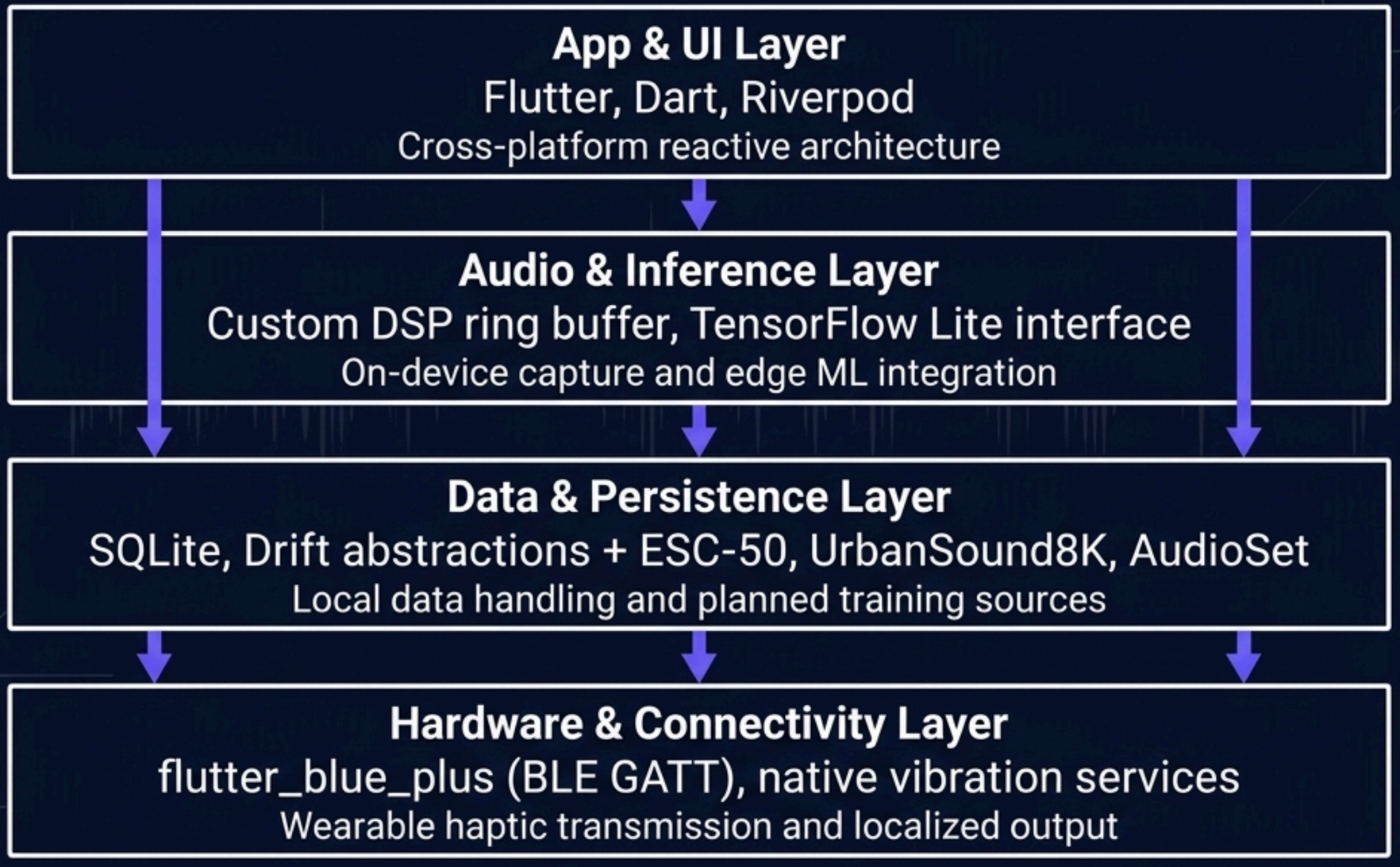


Threshold Mechanics:
- 2 to 5 recordings averaged into a taught-sound prototype.
- Default similarity threshold 0.85 (configurable 0.50 - 0.98).
- Adaptive threshold increase of +0.04 after 3 consecutive thumbs-down ratings.

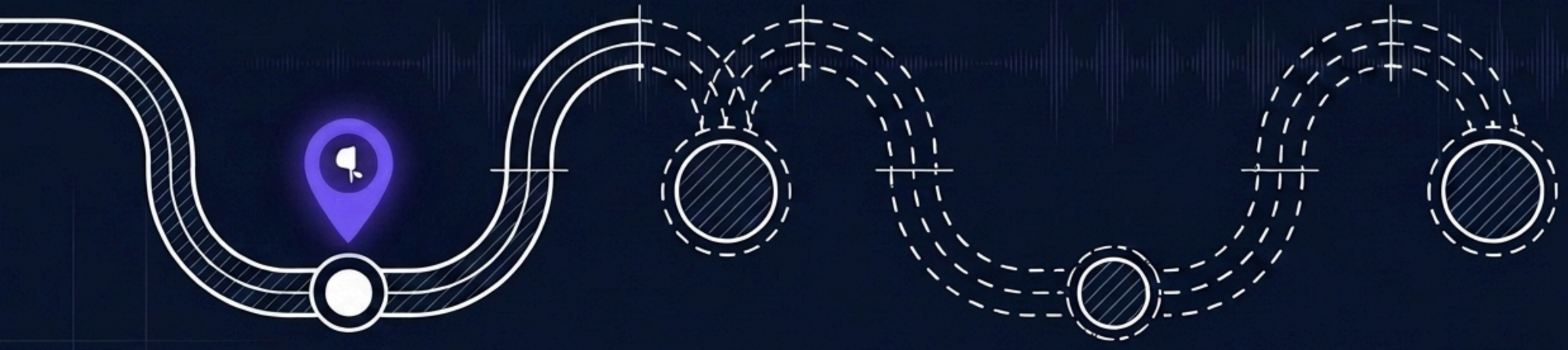
Methodology



Tools and Datasets



Work Plan



Review

Finalize architecture, software baseline modules, UI, and initial demonstration script.

Post Review

Prepare dataset mapping; execute transfer-learning pipeline; establish initial threshold metrics.

Mid Project

Integrate validated production model; run controlled tests on few-shot household sounds; refine debounce/alerts.

Final Review

BLE wearable hardware validation; gather usability feedback; finalize documentation and accuracy findings.

Initial Working Module Demonstration

Launch/Consent



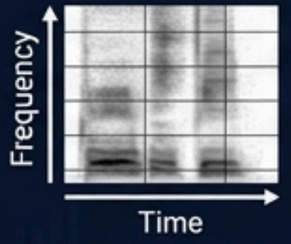
Onboarding, local-processing privacy validation.

Live Monitor



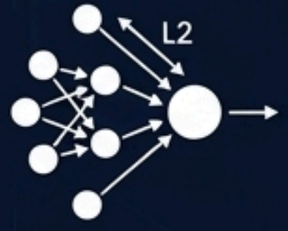
Waveform, dB meter, and status overlay.

DSP Diagnostic



Real-time log-mel spectrogram and category probabilities.

Teach a Sound



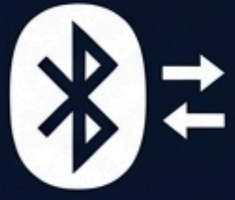
Capturing 2-5 samples and calculating L2-normalized embeddings.

Detection & Alert



10-second debounce test with haptic/visual strobe output.

BLE Mock Pairing



GATT haptic transmission to mock wearable.